

NAME: _____ PDG: _____ Register No: _____



ANDERSON JUNIOR COLLEGE

Preliminary Examinations 2011

CHEMISTRY

9647/03

Higher 2

15 September 2011

Paper 3 Free Response Questions

2 hours

Candidates answer on separate paper.

Additional Materials: Data Booklet
 Writing paper

READ THESE INSTRUCTIONS FIRST

Write your name, PDG and register number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

Start each question on a fresh sheet of paper.

You may use a pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer any **four** questions.

A Data Booklet is provided.

You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

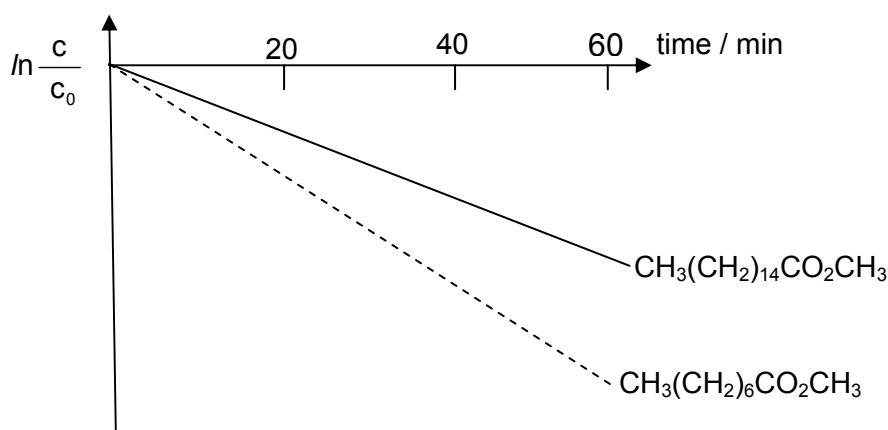
This document consists of **11** printed pages.

- 1 (a) Fatty acid methyl esters (FAME) have been widely used in the food and cosmetic industries. Methyl octanoate, $\text{CH}_3(\text{CH}_2)_6\text{CO}_2\text{CH}_3$, is an example of FAME.

To investigate the hydrolysis of methyl octanoate, $\text{CH}_3(\text{CH}_2)_6\text{CO}_2\text{CH}_3$, in subcritical water, a known concentration of $\text{CH}_3(\text{CH}_2)_6\text{CO}_2\text{CH}_3$ was dissolved in hexane. The dissolved $\text{CH}_3(\text{CH}_2)_6\text{CO}_2\text{CH}_3$ was then added to excess distilled water in a reactor resistant to high pressure and temperature. The kinetics of the reaction at 483 K was followed for 1 h.

- (i) Write a balanced equation to show the hydrolysis of methyl octanoate.
- (ii) In the above study, the distilled water was purged with helium gas before use. Suggest a reason for this.

In another investigation, the hydrolysis of methyl palmitate, $\text{CH}_3(\text{CH}_2)_{14}\text{CO}_2\text{CH}_3$, was studied at 483 K and 543 K. The following graphs were obtained for the experiment conducted at 483 K.



c : concentration of FAME at time t ;
 c_0 : concentration of FAME at time $t = 0$;

The rate constant for the hydrolysis of $6.32 \times 10^{-2} \text{ mol dm}^{-3}$ of $\text{CH}_3(\text{CH}_2)_6\text{CO}_2\text{CH}_3$ at 483 K was found to be 0.24 min^{-1} .

- (iii) Deduce the order of reaction with respect to $\text{CH}_3(\text{CH}_2)_6\text{CO}_2\text{CH}_3$.
- (iv) Sketch a graph of concentration of $\text{CH}_3(\text{CH}_2)_6\text{CO}_2\text{CH}_3$ against time at 483 K, including all essential details on your sketch.
- (v) With reference to the graphs given above, comment on the rates of hydrolysis of $\text{CH}_3(\text{CH}_2)_6\text{CO}_2\text{CH}_3$ and $\text{CH}_3(\text{CH}_2)_{14}\text{CO}_2\text{CH}_3$ at 483 K.

Hence predict the difference in their activation energies. Explain your reasoning.

- (vi) Copy the graph shown above for $\text{CH}_3(\text{CH}_2)_{14}\text{CO}_2\text{CH}_3$ at 483 K onto your writing paper. On the same axes, sketch the graph for $\text{CH}_3(\text{CH}_2)_{14}\text{CO}_2\text{CH}_3$ at 543 K. Label your graphs clearly.

Explain, in molecular terms, your answer.

[11]

(b) Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{H}$, is another common ingredient used in cosmetic products. In nature, (–)-Lactic acid is produced from the fermentation of lactose. Lactose exists in two forms, the α -form which has a specific rotation of $+92.6^\circ$ and the β -form which has a specific rotation of $+34^\circ$ at 20°C .

- (i)** If an aqueous solution containing both forms of lactose is allowed to stand, the optical rotation changes until it reaches $+52.3^\circ$.

Determine the percentage composition of the α -form in the equilibrium mixture.

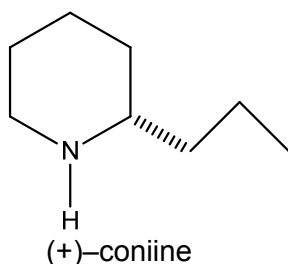
- (ii)** Lactic acid can also be chemically synthesized from ethanal, CH_3CHO in a 2-stage synthesis.

Suggest reagents and conditions for each step and draw the structural formula of the intermediate compound.

- (iii)** Briefly outline the mechanism of the reaction involved in the first step of the synthesis in **(ii)**.

With reference to your mechanism, explain why lactic acid synthesized by this method is optically inactive.

- (iv)** The optically inactive lactic acid obtained in **(iii)** is reacted with (+)-coniine.



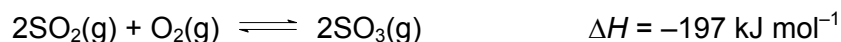
Draw a displayed formula of one of the products formed, indicating each chiral centre with a (*).

[9]

[Total: 20]

2 This question involves sulfur and its compounds

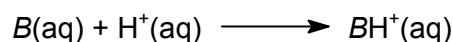
- (a) The key step in the manufacture of sulfuric acid is the reaction between sulfur dioxide and oxygen in excess air.



- (i) For the above reaction, write an expression for the equilibrium constant, K_p , stating the units.
- (ii) Explain why excess air is required in the reacting gas mixture.
- (iii) Describe and explain the effect of increase in temperature on the
 - I. rate at which equilibrium is achieved.
 - II. equilibrium constant K_p .

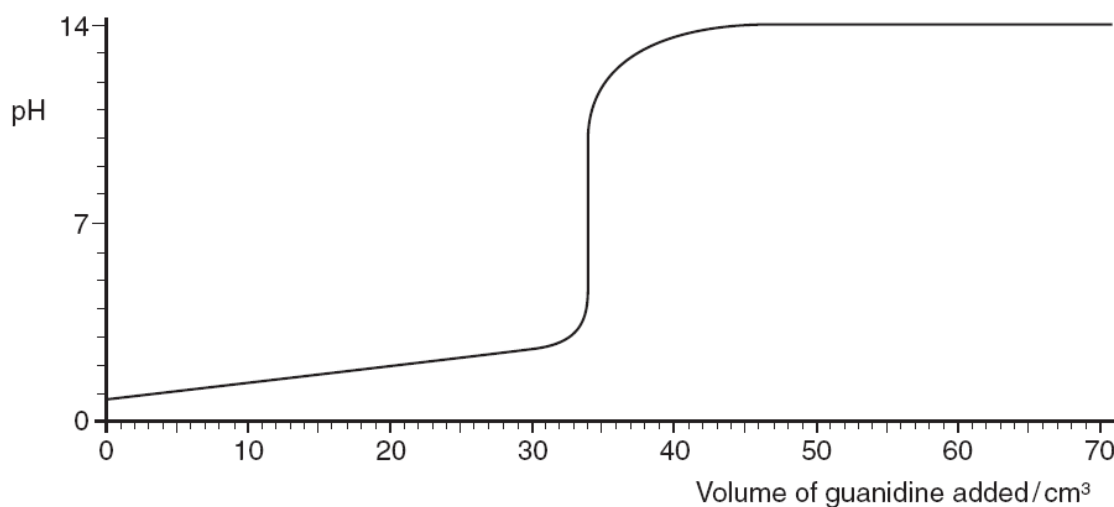
[6]

- (b) An organic base, guanidine, reacts with sulfuric acid as follows:



where B represents the molecule of guanidine.

When a 25.0 cm^3 sample of dilute sulfuric acid was titrated against a solution of guanidine, the following titration curve was obtained.



- (i) Is guanidine a strong or weak base? Explain your answer.
- (ii) The initial pH of the sulfuric acid is 0.70.
Calculate $[\text{H}^+]$, and hence the initial concentration of sulfuric acid.
- (iii) Calculate the concentration, in mol dm^{-3} , of guanidine in the solution.
- (iv) The guanidine solution contained 8.68 g of the base per dm^3 .

Use your answer to (iii) to calculate the M_r of guanidine.

[6]

- (c) One of the major industrial uses of sulfuric acid is to convert phosphate rock into dihydrogen phosphates (H_2PO_4^-), which makes useful buffers for biochemical experiments.



- (i) Calculate the pH at which maximum buffering capacity occurs for a solution that contains NaH_2PO_4 and Na_2HPO_4 .
- (ii) Hence, suggest a suitable indicator for the titration of H_2PO_4^- with aqueous sodium hydroxide. Explain your answer.
- (iii) Hydrogen phosphate, HPO_4^{2-} , can behave both as an acid and a base. Using the given K_a values, explain why it is a stronger base than it is an acid.
- (d) Sodium sulfate is often given as a first aid for barium poisoning as barium sulfate will be precipitated. The numerical value of the solubility product, K_{sp} , for barium sulfate is 1.08×10^{-10} .

- (i) Write an expression for K_{sp} of barium sulfate, stating the units.
- (ii) Calculate the concentration of a saturated barium sulfate solution.
- (iii) Explain whether barium sulfate will be more or less soluble in water compared to in a 0.1 mol dm^{-3} barium hydroxide solution.

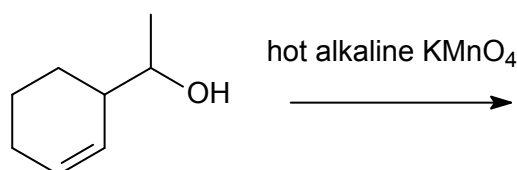
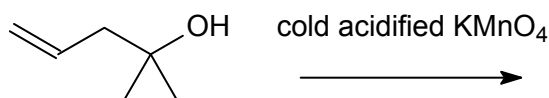
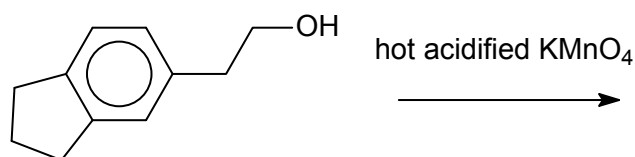
[3]

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- 3 Selective oxidation of alcohols to aldehydes and ketones is a key reaction in organic synthesis, both for fundamental research and industrial manufacturing.

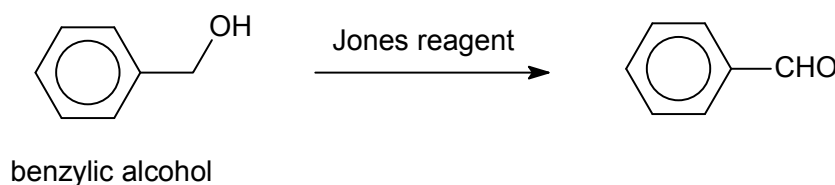
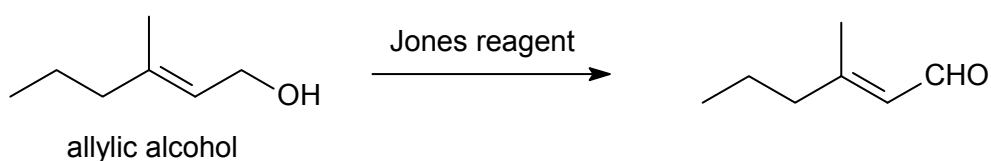
(a) Under controlled conditions, an aqueous solution of potassium manganate(VII), KMnO_4 oxidises primary alcohols to carboxylic acids very efficiently. When alkaline $\text{KMnO}_4(\text{aq})$ is added to a primary alcohol, the purple solution first becomes dark green and then produces a brown precipitate.

- State the identities of the dark green solution and the brown precipitate.
- Describe and explain the colour change that will be observed if acidified $\text{KMnO}_4(\text{aq})$ is added to the primary alcohol instead.
- Predict the organic products of the following reactions with $\text{KMnO}_4(\text{aq})$:

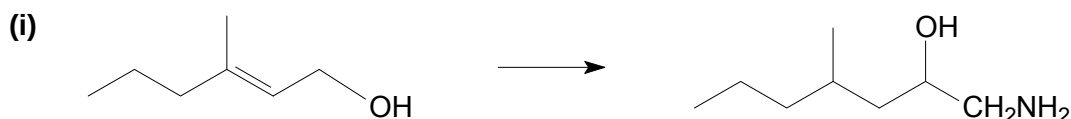


[7]

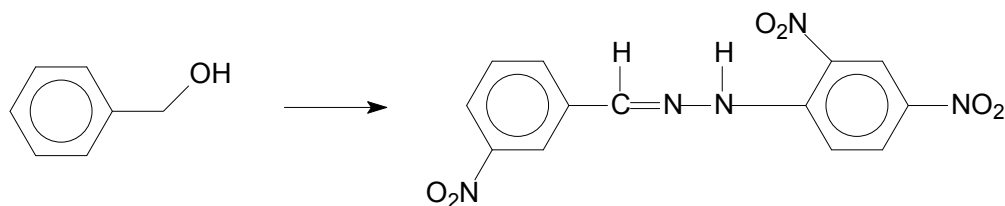
- (b) In 1975, Kenn E. Harding and his co-workers published their findings on the use of Jones reagent in the selective oxidation of allylic and benzylic alcohols to the corresponding aldehydes. Two such reactions are shown below.



By making use of the above oxidation reactions, devise a synthetic pathway (in not more than 3 steps) for the following conversions:



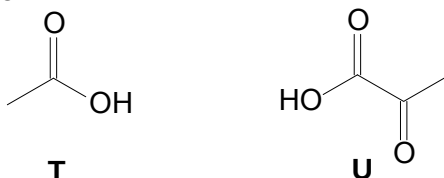
(ii)



In your answer, state the reagents and conditions required for each step as well as the structure of any intermediates formed.

[6]

- (c) Compounds **P** and **Q**, with molecular formula $C_7H_{12}O$, are isomers of each other. 1 mole of each compound, **P** and **Q**, reacts with 2 moles of hydrogen gas in the presence of a platinum catalyst. On treatment with Jones reagent, **P** and **Q** produce **R** and **S** respectively, both with molecular formula $C_7H_{10}O$. Both **R** and **S** give an orange precipitate on reaction with 2,4-dinitrophenylhydrazine. **R** gives a silver mirror with Tollens' reagent but not **S**. When **R** and **S** are added separately to hot acidified potassium manganate(VII), the same compounds **T** and **U** are obtained as the only organic products. The structures of **T** and **U** are given below:

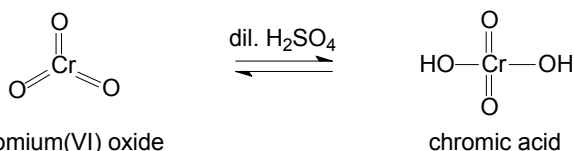


Hence, suggest the structural formulae of compounds **P** – **S**.

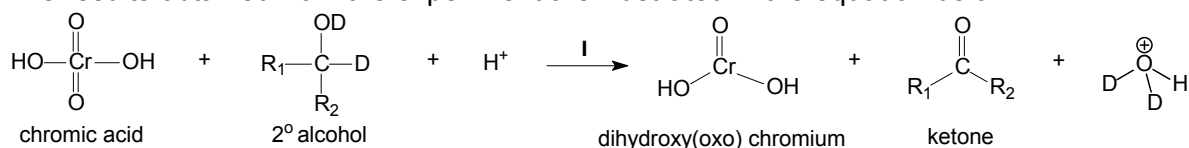
[4]

- (d) Jones reagent is also often used in the oxidation of secondary alcohols to ketones. An isotopic labeling experiment involving the use of deuterium, D, an isotope of hydrogen, can be performed to study this oxidation mechanism.

The Jones reagent is a solution of chromium(VI) oxide, CrO_3 in dilute sulfuric acid, which forms chromic acid in situ as shown below.



The results obtained from the experiment are illustrated in the equation below.



- (i) Determine the change in the oxidation number of Cr in chromic acid and dihydroxy(oxo) chromium.
- (ii) An approximate value for ΔH of reaction **I** can be calculated using bond energy values. Given the following bond energy data, and using other appropriate bond energies from the *Data Booklet*, calculate the value of ΔH for reaction **I**.

bond	bond energy/kJ mol ⁻¹
C–D	340
Cr=O	478
O–D	430

[3]

[Total: 20]

- 4 Magnesium is seldom found naturally in its pure state as a metal. It readily reacts with other elements to form a large number of compounds, which have different commercial uses, depending on their properties. For example, magnesium carbonate has a high lattice energy and can be used to produce bricks for furnaces.

(a) (i) Explain the term *lattice energy*.

- (ii) Use relevant data from the *Data Booklet* and the following experimental data to construct an appropriate energy cycle and calculate the lattice energy of magnesium carbonate.

enthalpy change of atomisation of magnesium = $+148 \text{ kJ mol}^{-1}$

sum of first and second electron affinities of oxygen = $+657 \text{ kJ mol}^{-1}$

enthalpy change of formation of magnesium carbonate = $-1096 \text{ kJ mol}^{-1}$

enthalpy change of formation of $\text{CO}_2(\text{g}) = -393 \text{ kJ mol}^{-1}$

$\text{CO}_3^{2-}(\text{g}) \longrightarrow \text{O}^{2-}(\text{g}) + \text{CO}_2(\text{g}); \Delta H = +778 \text{ kJ mol}^{-1}$

- (iii) The theoretical value for the lattice energy of magnesium carbonate is $-2350 \text{ kJ mol}^{-1}$. Comment on the difference in this value and the experimental value you found in (ii).

- (iv) Use your answer and relevant data in (ii) to calculate the enthalpy change of decomposition of magnesium carbonate, given that the lattice energy of magnesium oxide is $-3800 \text{ kJ mol}^{-1}$.

[7]

- (b) The mineral dolomite is a double carbonate of magnesium and calcium, with the formula $\text{CaMg}(\text{CO}_3)_2$. The product resulting from its decomposition can be used as a refractory material.

Below 900°C , dolomite decomposes partially to form a solid mixture. It was found that for every mole of dolomite that was partially decomposed, 1 mol of gas was produced.

- (i) Identify the products of the decomposition of dolomite below 900°C .

- (ii) Explain the formation of the products in (i).

[3]

- (c) Above 900°C , dolomite decomposes completely. The resulting solid mixture is highly porous and reactive and is known as 'calcinated dolomite'. When a certain mass of dolomite was decomposed completely, 2142 kJ of energy was absorbed and 700 g of 'calcinated dolomite' was formed.

- (i) Write an equation to represent the formation of 'calcinated dolomite'.

- (ii) Use your answer in (a)(iv) to calculate a possible value for the enthalpy change of decomposition of calcium carbonate.

[3]

(d) Magnesium chloride, MgCl_2 , can be contained in supplements used to treat magnesium deficiency in the body. It dissolves in the water in the blood, which is maintained at a normal pH of 7.4, and magnesium is absorbed by the body in ionic form.

(i) Describe the reaction of magnesium chloride with water and write equations where appropriate.

(ii) Phosphorous can be contained in supplements to treat phosphorous deficiency in the body as well.

Explain, with the use of an equation, whether phosphorous(V) chloride is likely to be contained in such supplements.

(iii) In the laboratory used for producing magnesium supplements, magnesium chloride and magnesium bromide were mixed up.

Explain how they can be differentiated from the products formed in their reactions with hot concentrated sulfuric acid and suggest a reason for the difference in the observations.

[7]

[Total: 20]

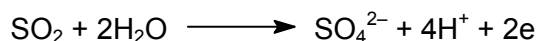
- 5 Vanadium is a transition metal that can form stable coloured ions of various oxidation states in aqueous solutions. Some of the ions of vanadium and their corresponding colours are shown in the table below.

formula of vanadium ion	VO_3^-	VO^{2+}	V^{3+}	V^{2+}
colour of aqueous solution	yellow	blue	green	violet

- (a) (i) Explain why aqueous solutions of vanadium ions are coloured.
- (ii) Use E° values from the *Data Booklet* to predict what will be formed when zinc is added to an aqueous solution of ammonium vanadate(V), NH_4VO_3 .

Hence state the colour of the final solution and write balanced equations for any reactions that occur.

- (iii) When 3.00 g of a sample of NH_4VO_3 was reacted with sulfur dioxide, a blue solution was obtained. In this reaction, sulfur dioxide is oxidised according to the equation given below.



Calculate the volume of sulfur dioxide that is required to reduce the VO_3^- ions at room temperature and pressure.

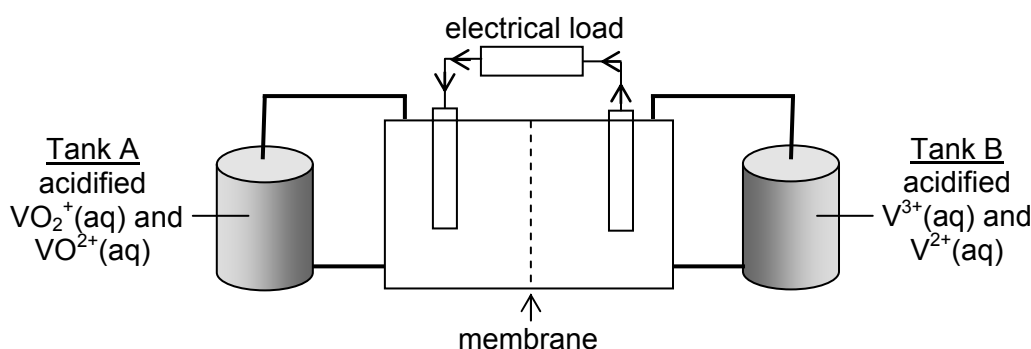
[8]

- (b) The dioxovanadium(V) ion, VO_2^+ , can act as a homogeneous catalyst for the reaction between iodide ions, I^- and peroxodisulfate ions, $\text{S}_2\text{O}_8^{2-}$.

- (i) What do you understand by the term *homogeneous catalyst*?
- (ii) By considering suitable E° values from the *Data Booklet*, explain how VO_2^+ functions as a catalyst for the reaction between I^- and $\text{S}_2\text{O}_8^{2-}$, writing equations where appropriate.

[3]

- (c) One important commercial use of vanadium compounds is the vanadium redox battery. This rechargeable battery is a flow battery with graphite electrodes, where the electrolytes can be pumped from separate storage tanks into the cell. A thin membrane, separating the two electrolytes, allows selected ions to flow through it.



The direction of the electron flow during the discharging process is indicated in the diagram of the battery above.

- (i) Write the equations for the reactions that occur at the cathode and the anode.

- (ii) Calculate the e.m.f. of the cell, assuming standard condition.
- (iii) Why is it necessary to have a membrane that allows selected ions to flow through it in the vanadium redox battery?
- [4]
- (d) The battery supplies a continuous current of 14 A for 20 hours. During discharge, VO_2^+ is converted to VO^{2+} .
Given the volume of the electrolyte in the cell is 5 dm^3 , calculate the minimum initial concentration of the VO_2^+ required.
- [2]
- (e) The electrolyte in Tank B and its corresponding electrode in the vanadium redox battery are modified in an attempt to increase the e.m.f. of the cell.
- (i) Using the *Data Booklet*, choose an appropriate electrode and electrolyte that will increase the e.m.f of the battery.
- Hence calculate the e.m.f. of the modified cell, assuming standard conditions.
- (ii) State one disadvantage of the modified cell you have chosen in (i) compared to the conventional vanadium redox battery.

[3]
[Total: 20]